

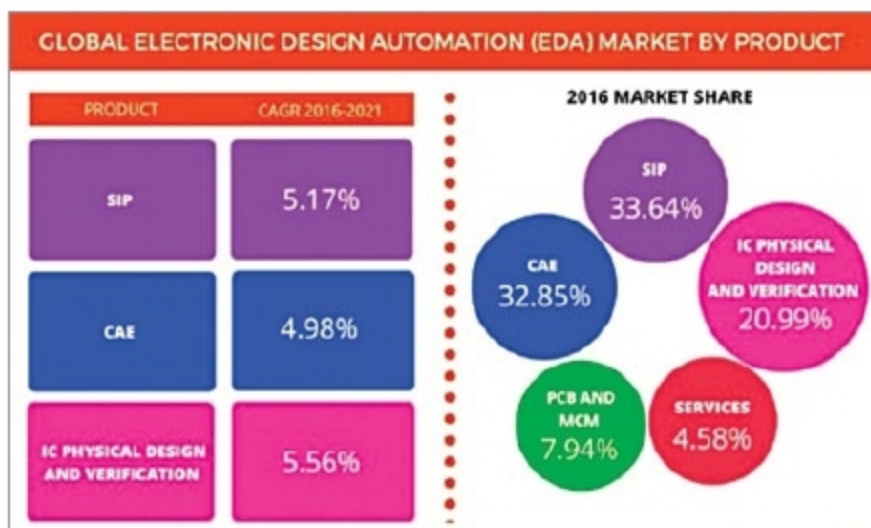


Fig. 2: Global EDA market by products (Source: Technavio)

tions is likely to be the biggest driver of growth of the electronics design market. In addition, there are a few emerging technologies still in their nascent stage, which could emerge as key areas of focus in the near future. These include the following:

- Next-generation wireless
- AI
- Silicon photonics

India has become the hub for electronics design with nearly 2000 chips being designed per year and around 500,000 engineers working in various aspects of electronic design, starting from chip design and verification to different stages of product design. Though still lagging behind chief competitors in the far east (that is, China and Taiwan), India has significant potential as it seeks to continually raise the bar, and increase the breadth and depth of its offerings.

Within semiconductor design, there is very large-scale integration (VLSI) design, embedded software and electronic design automation (EDA). In India, VLSI and EDA hold the biggest chunk in the Indian semiconductor design industry. While VLSI is big in Bengaluru, Noida holds the biggest chunk of EDA operations in India. This analysis categorises the global EDA market into five major product segments. These are:

- Semiconductor intellectual property (SIP)
- Computer-aided engineering (CAE)
- IC physical design and verification
- Printed circuit board (PCB) and multi-chip module (MCM)
- Services

integration have compelled semiconductor manufacturing companies to use the available EDA tools.

EDA tools help reduce the overall product development time from two to three years, to two to three weeks. Continuous growth in the semiconductor industry has triggered an increase in the demand for EDA tools. The Indian electronics design market is expected to be driven by this global trend.

Greater digitisation across segments lead to increased penetration of Indian electronics design houses in such application areas as LED illumination, communications and

According to a study released by Technavio, the global electronics design automation market is expected to grow at a CAGR of over five per cent during the period 2017-2021. According to that report, persistent decrease in the size of semiconductor chips and increased scale of

broadband equipment, and automotive electronics. Localisation of technological changes along with global competitiveness prove that the world is considering India as the design house for the global electronics industry.

India's strength in design systems is based on three main factors, as given below.

- Strong capabilities in embedded system design
- Large number of electronics engineering graduates (even though the useful percentage of those who can be employed is still around 10 per cent)
- Increased demand due to a sizeable number of Indian SMEs and startups focusing on the IoT

Market challenges

In recent years, however, independent design houses in India are growing in stature and are increasingly involved in developing designs for global firms. While conditions are favourable for the growth of the industry, there are a few impediments that need to be removed for the industry to live up to the expectations.

There are very few Indian companies that are capable of doing the complete product design. The reason being lack of skill-sets at various levels. For this, one needs to possess expertise to understand and define overall architecture, to translate it into hardware and software requirements, and to do the actual implementation. There are not too many Indian companies that have in-depth expertise in ASIC/FPGA design, high-speed board development and firmware development.

The Indian electronics industry is still more focussed on assembly than manufacturing. Therefore semi-finished items are imported from China and only the final assembly is done in India. For example, cellphones and smart TVs are designed in China and assembled in India. Thus, only a small percentage of design for local brands happens here.

One of the prime reasons for this is the requirement of significant investment in R&D in a sector where

The Indian electronics design market

Major types of services are:

- Embedded software design
- End-to-end product design
- EDA
- VLSI design
- Product certification and compliance

Major types of companies include:

- Independent design houses offering design services for the world market more in the services business model (driven by the IoT, automation, analytics, etc) and designing India-specific products (especially with the IoT and automation driving efficiencies for manufacturing, healthcare and hospitality segments)
- Captive design houses of global electronics firms
- EDA tool providers
- Original design manufacturers